

HAYAT MEDICAL COLLEGE

Progressive assessment II for Year I Medical Students

Date: December 19, 2014

Time allowed: 3:00 hours.

Name: _____

HISTOLOGY

MCQ: Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case. (Total : 50 points)

_____1. Concerning the **teeth and associated tissues**, which statement is **true**?

- a) Dentin is more calcified than enamel.
- b) Pulp is composed of dense connective tissue.
- c) Enamel is replaced in the root by cementum.
- d) Odontoblasts make enamel.
- e) None of the above.

_____2. Which of the following statements is **false** in the **oral cavity**?

- a. Taste buds occur most commonly on the sides of fungiform and circumvallate papillae.
- b. The whitish tips of filiform papillae consist of keratinized epithelial tissue.
- c. Sublingual gland lacks serous acini but contains serous demilunes.
- d. Periodontal ligament is composed of mucoid connective tissue.
- e. Serous demilunes produce lysozyme.

_____3. Regarding the **stomach**, which statement is **true**?

- a) Is able to withstand the acidic environment because it is lined by stratified squamous epithelium.
- b) Has its gastric glands best developed in the pylorus than the other regions.
- c) Forms temporary folds called plicae circulares when it is empty.
- d) Has its muscularis externa made by skeletal muscle fibres.
- e) Mucosal glands of the pylorus secrete mainly mucus.

_____4. Concerning the **stomach**, which statement is **true**?

- a. It is specifically responsible for the absorption of vitamin B₁₂ and the reabsorption of conjugated bile salts.
- b. Goblet cells are more abundant to produce mucus as well as appreciable amounts of lysozyme.
- d. Hydrochloric acid and intrinsic factors are secreted by enteroendocrine cells of cardia and pylorus.
- d. Parietal cells also called oxyntic cells contain mucinogen granules and pepsinogen.
- e. The folds of the empty stomach are called rugae.

_____5. Which one of the following marks best the **ileum** from the **duodenum** and **jejunum**?

- a) Long and broad leaf like villi.
- b) Absence of taenia coli.
- c) Peyer's patches.
- d) Plicae circulares.
- e) Well developed intestinal glands.

_____6. With regard to the **small intestine**, which statement is **true**?

- a. The muscularis externa in the upper third of duodenum and jejunum is composed of striated muscle.
- b. Crypts of Lieberkühn are lymphatic capillaries [lacteals] within a villus of the small intestine.
- c. Ileum characteristically contains the most goblet cells in the mucosa covering its villi.
- d. Plicae circulares are projections with a core of mucosa.
- e. Villi are finger like projections of submucosa.

- 7. **Diarrhea** may result if which of the following organs fails to carry out its primary role in **absorbing water from the feces**?
- a) anal canal b) colon c) cecum d) small intestine e) stomach
- 8. With regard to the **large intestine**, which statement is **false**?
- a. Lacks villi and contains a disproportionately large number of goblet cells.
b. The large intestine is shorter than the small intestine but has a wide lumen
c. The muscularis externa has three thick bands of muscle called the tenia coli.
d. The lamina propria has many macrophages, plasma cells, eosinophils, and lymphoid nodules.
e. Paneth cells of the submucosal glands house bacteria that produce vitamin B₁₂ and vitamin K.
- 9. Which of the following statements is **correct** regarding the **intestine**?
- a) The striated border of the small intestine is made by microvilli. b) Small intestine has tenia coli.
c) The absorptive cells of the small intestine undergo mitosis at the tips of the villi and are sloughed into the crypts. d) Large intestine possesses branched alveolar glands.
e) Brünner's glands in mucosa of the duodenum produce mucin containing alkaline fluid and the polypeptide hormone urogastrone and empty their secretions into the lacteals.
- 10. Concerning the **Vermiform appendix**, select the **correct** statement.
- a) has few lymphoid nodules in the lamina propria and submucosa.
b) is a blind-ended evagination of the jejunum. c) has relatively shallow intestinal glands.
d) longitudinal smooth muscle gathers into its teniae coli. e) contains short villi.
- 11. Which is **not the function of the liver**?
- a. Storage of glycogen, fat soluble vitamins, iron and cholesterol.
b. Detoxification/removal of drugs and alcohol. c. Storage of calcium and bile.
d. Production of plasma proteins and lipoproteins. e. Fetal RBC production.
- 12. With regard to the **pancreas**, which statement is **false**?
- a. Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
b. Myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
c. The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
d. It is a lobulated retroperitoneal gland whose islet cells are compact mass of epithelial cells not cords or plates.
e. Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
- 13. Which structure has a **mucosa lined by nonkeratinized stratified squamous epithelium**?
- a. Esophagus b. Stomach c. Small intestine d. Large intestine e. Rectum
- 14. Concerning the **GI tract**, select the statement which is **not correct**.
- a. The muscularis externa consists of Auerbach's plexus.
b. Hepatic sinusoids are surrounded by the space of Disse.
c. Masticatory mucosa is located on the hard palate and gingiva.
d. The mucosa of the gallbladder is made of simple columnar epithelium.
e. The adventitia consists of only a single layer of epithelial cells and underlying connective tissue.
- 15. With regard to the **GI tract**, select the statement which is **correct**.
- a. Permanent circular folds (plicae circulares) are characteristic structural features of the esophagus.
b. Fat-storing stellate cells (Ito cells) in the space of Disse in the liver are noted for storing vitamin A.
c. Myenteric (Auerbach's) plexus is located just beneath the epithelium of the mucosa, in lamina propria.
d. A hepatic cord is the tough connective tissue surrounding a portal triad.
e. Like the small intestine, the gall bladder has villi with lacteals adjacent to the Paneth and goblet cells in the mucosal epithelium.

————16. **Transitional epithelium** (urothelium) lines all of the following **EXCEPT** the:
a. Bowman's capsule. b. renal pelvis. c. ureter. d. bladder. e. proximal urethra in males.

————17. Which of the following is characteristic of the **proximal** but **not** of the **distal convoluted tubules**?
a) convoluted portion lies in the renal cortex b) epithelial lining is similar to that of the loop of Henle
c) exhibits a brush border d) lining cells are basophilic
e) secretes glucose and amino acids into the filtrate

————18. Which of the following is **not** a **function of the kidney**?
a. Erythropoietin production b. Vitamin D modification c. Acid base balance
d. Aldosterone production e. Renin production

————19. Which of the following is **not** the function of **mesangial cells**?
a. synthesis of extracellular matrix. b. endocytosis and contractility.
c. maintenance of acid-base balance in the blood. d. structural support to the glomerulus.
e. increase blood flow and the effective surface area available for ultrafiltration.

————20. In the **urinary system**, which statement is **false**?
a. Aldosterone acts on the distal convoluted duct.
b. Peritubular capillaries arise from the efferent arterioles.
c. The renal pyramid and its associated cortex is referred to as the renal lobe.
d. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
e. Renin is secreted by podocytes of the visceral layer of Bowman's capsule referred to as area cribrosa.

————21. In the **urinary system**, which statement is **true**?
a. Interstitial cells in the renal medulla secrete erythropoietin through filtration slits.
b. Macula densa are closely-packed nuclei in the epithelium of the proximal convoluted tubule.
c. Endothelial cells of glomerular capillaries are of non-fenestrated variety.
d. The distal convoluted tubules possess neither brush borders nor apical canaliculi.
e. The cortical collecting tubules are located primarily at the apex of the renal pyramids and are lined by pseudostratified ciliated columnar cells.

————22. Epithelial cells lining the **thin limb of the loop of Henle**:
a. are modified smooth muscle cells responsible for the secretion of rennin.
b. have an apical brush border of microvilli for active absorption from the lumen.
c. contain many ribosomes and rough endoplasmic reticulum for extensive protein synthesis.
d. contain numerous basally located secretory vesicles. e. are simple squamous.

————23. **Testis** has the following coverings **except**:
a. tunica adventitia b. tunica vasculosa c. tunica albuginea d. tunica vaginalis e. None

————24. In the **testis**, which statement is **true**?
a. The seminiferous tubules are lined with a complex stratified epithelium.
b. The secondary spermatocytes are the largest cells of the spermatogenetic lineage.
c. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
d. Interstitial cells (of Leydig) produce testicular fluid which constitute the bulk of ejaculate volume.

e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

—————25. Which of the following is **NOT** considered part of the **male genital duct system**?

- a. Rete testis b. Tubuli recti c. Seminal vesicles d. Ductus deferens e. Ductus epididymidis

—————26. Which of the following statements is **false** in the **male reproductive system**?

- a. The secretion of seminal vesicle is particularly rich in fructose.
b. Secondary spermatocytes are smaller than the primary spermatocytes.
c. The epithelium lining the ductus epididymis is pseudostratified columnar.
d. Tunica albuginea is the thickest in epididymis. e. The subcutaneous tissue in penis is devoid of fat.

—————27. The function of **Sertoli cells** includes all of the following **EXCEPT**:

- a. phagocytosis of residual bodies from maturing spermatids.
b. establishment through tight junctions of a blood/testis barrier. c. secretion of testosterone.
d. metabolic and mechanical support of germ cells. e. secretion of tubular fluid.

—————28. Which of the following statements is **true** with regard to the **male reproductive system**?

- a. A network of tubules within the testis which interconnects the seminiferous tubules with the efferent ductules is the epididymis.
b. Epithelial lining of the duct of epididymis is columnar epithelium with brush borders and goblet cells.
c. The ejaculatory duct connects the epididymis with the seminal vesicle where sperm cells mature after spermiogenesis.
d. Semen contains abundant fluid derived from the lymphatics lined by mesothelium of the corpus spongiosum and corpora cavernosa.
e. The seminal vesicle consists of a hollow tube with an extensively folded mucosal lining, in a fibromuscular stroma.

—————29. Which of the following statements is **false** in the **male reproductive system**?

- a. The penile urethra passes through the corpus spongiosum.
b. The secretion of seminal vesicle is particularly rich in fructose.
c. The prostate gland, the largest accessory structure, is composed of many compound tubuloalveolar glands.
d. Ductus epididymidis secretes glycerophosphocholine. e. None of the above.

—————30. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?

- a) Contains mucus-secreting peg cells.
b) Contains large mucosal folds that are extensively branched. c) Contains smooth muscle in its walls.
d) Is covered by serosa derived from the peritoneum. e) Contains ciliated epithelial cells.

—————31. Which of the following statements is **true** about the **female reproductive system**?

- a. Mucosa of the uterus is commonly called perimetrium.
b. Fornix of vagina is lined by stratified squamous epithelium.
c. The life span of corpus luteum of pregnancy is about 10 lunar months.
d. Primary oocyte completes its first meiotic division at the time of fertilization.
e. The basic pH in the vagina is due to the secretion of HCl by the tubuloalveolar glands of the vaginal walls.

—————32. Which of the following statements is **true** in the **female reproductive system**?

- a. The cells of theca interna secrete progesterone.
b. Germinal (surface) epithelium covering the ovary gives rise to ova.

- c. Glassy membrane is the last structure to disappear in an atretic follicle.
- d. Corpora albicans are ovarian follicles which degenerate after failing to mature.
- e. After ovulation, granulosa cells and cells of theca interna grow and differentiate to form the primordial follicles.

.———33. In the **ovary**, which statement is **true**?

- a. During ovarian follicle maturation, surrounding stromal cells differentiate to form the zona pellucida.
- b. Ovarian follicles which degenerate after failing to mature are called "*white bodies*" or corpora albicantia.
- c. Primary oocyte completes its first meiotic division at the time of ovulation.
- d. Primordial follicle is identified in section by the presence of cumulus oophorus.

.———34. **Liquor folliculi**,

- a) stimulates the myoepithelial cells of the mature follicle to rupture.
- b) acts as a porous filter through which dissolved hormones and enzymes are prevented.
- c) causes the endometrial glands to secrete and prepare the endometrium for implantation.
- d) is composed of cumulus oophorus and polar bodies.
- e) None of the above.

———90. The **proliferative phase of the endometrium** is characterized by all of the following **EXCEPT**:

- a. common mitotic figures in the endometrial stroma.
- b. common mitotic figures in the glandular epithelium.
- c. absence of differentiated secretory cells in the glandular epithelium.
- d. relatively smooth epithelial lining of the glandular tubules.

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- e. a stratified squamous surface.

Cilia are found on the free surface of corona radiata cells. xxxx

Mammary glands are simple tubuloacinar glands, special modified sebaceous glands.

xxxx

Uterine tube is lined by simple columnar ciliated epithelium.

Nonciliated secretory cells present in the lining epithelium of uterine tube are called peg cells.

———11. In the **ovaries**, which statement is true?

- a) The capsule contains tunica albuginea of dense irregular collagenous connective tissue.
- b) The primary follicles are dependent on FSH for their development.
- c) Corpus albicans is a temporary endocrine gland whose formation depends upon LH surge.
- d) Atretic follicles are found in the ovarian medulla.
- e) Theca interna cells manufacture androgens.

54. In the female reproductive system, which of the following statements is false?

- a. Ovulation is triggered by a dramatic increase in LH.
- b. During a pregnancy, hCG maintains the corpus luteum.
- c. Proliferative phase of uterus coincides with the development of ovarian follicles.
- d. In the ovaries, the medulla harbors most of the oocyte-containing ovarian follicles.
- e. during pregnancy the endometrium grows extensively by hypertrophy and hyperplasia.

55. Which of the following is true of the uterine tube's ampulla but not of its isthmus?

- a) contains mucus-secreting peg cells
- b) contains ciliated epithelial cells
- c) is covered by serosa derived from the peritoneum
- d) contains smooth muscle in its walls
- e) contains large mucosal folds that are extensively branched

56. Which of the following statements is false in the female reproductive system?

- a. Mammary gland is a modified apocrine sweat gland.
- b. The vagina lacks glands throughout its length and is lubricated by secretions from cervical glands.
- c. Lactating mammary gland secretes progesterone.
- d. Cervix does not participate in the endometrial (menstrual) phases.
- e. None of the above.

3. Which stage of the follicle is arrested in prophase?

- a. Primordial follicle
- b. Primary follicle
- c. Secondary follicle
- d. Mature follicle
- e. Graffian follicle

4. What is the cavity within a secondary follicle?

- a. Graffian follicle
- b. Theca folliculi
- c. Granulosa cells
- d. Zona pellucida
- e. Antrum

During the first week to 10 days, FSH is the main hormone which stimulates the growth of the follicles. Ovulation is triggered by a dramatic increase in LH.

During a pregnancy, HCG is the hormone which maintains the corpus luteum.

7. Which stage of the follicle is marked by the surrounding of flattened (squamous) follicular cells becoming cuboidal?

- a. Primordial follicle
- b. Primary follicle
- c. Secondary follicle
- d. Mature follicle
- e. Graffian follicle

The connective tissue layer around the primary follicle is the theca folliculi.

The acidophilic glycoprotein coat surrounding the oocyte is the zona pellucida.

A cavity within the follicle is the antrum.

Underneath the germinal epithelium of the ovary is connective tissue called the tunica albuginea.

During a pregnancy, hCG is the hormone which maintains the corpus luteum.

12. Which follicular stage is also called an antral follicle?

- a. Primordial follicle
- b. Primary follicle
- c. Secondary follicle
- d. Mature follicle
- e. Graffian follicle

2. What is the acidophilic glycoprotein coat which surrounds the oocyte?
a. Graffian follicle b. Theca folliculi c. Granulosa cells d. Zona pellucida
e. Antrum

5. Which stage of the follicle is characterized by a surrounding of flattened (squamous) follicular cells?

- a. Primordial follicle b. Primary follicle c. Secondary follicle
d. Mature follicle e. Graffian follicle

8.

What is the connective tissue layer around the primary follicle?

- a. Graffian follicle b. Theca folliculi c. Granulosa cells
d. Zona pellucid e. Antrum

10. What is the connective tissue which surrounds the ovary?

- a. Follicle b. Germinal epithelium c. Medulla d. Tunica albuginea
e. Cortex

_____ 1. Concerning *oogenesis*, which statement is true?

- a) All oogonia become primary oocytes during ovulation.
b) Primary oocytes begin the first meiotic division before birth.
c) Oogonia grow and give rise to secondary oocytes by mitotic division.
d) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
e) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.

_____ 2. Concerning *spermatogenesis*, one statement is false.

- a) Type A spermatogonia undergo mitosis to provide a continuous supply of stem cells throughout the reproductive life of the male.
b) Primary spermatocytes in males begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.
c) The primary spermatocytes are the largest germ cells in the seminiferous tubules.
d) No primary spermatocytes form after birth. e) None of the above.

_____ 3. With regard to *spermatogenesis*, which statement is false?

- a. spermatids are transformed into spermatozoa by spermiogenesis.
b. The entire process of spermatogenesis takes approximately 72 hours.
c. Spermatogenesis is regulated by LH production by the pituitary.
d. Spermatogenesis begins at puberty.
e. Spermatogonia begin to increase in number at puberty.

_____ 4. With regard to *Sertoli cells*, which statement is false?

- a. They are involved in the regulation of spermatogenesis.
b. They line the seminiferous tubules c. They support and nurture the male germ cells.
d. They coagulate semen and form a vaginal plug e. They produce testicular fluid.

- _____ 5. Regarding the *sperm transport*, which statement is false?
- a) Vesiculase coagulates some of the semen and forms a vaginal plug.
 - b) Fructose in the semen, secreted by the seminal glands, is an energy source for the sperms.
 - c) Spermatozoa are nonmotile during storage in the epididymis, but become motile in the ejaculate.
 - d) Prostaglandins in the semen stimulate uterine motility.
 - e) None of the above.

- _____ 6. When does a *secondary oocyte* complete its second meiotic division to become a mature ovum?
- a) at ovulation
 - b) before ovulation
 - c) at fertilization
 - d) at puberty
 - e) before birth

- _____ 7. Which statement is true about *female reproductive cycles*?
- a) LH stimulates the development of ovarian follicles and the production of estrogen by its follicular cells.
 - b) ovulation is triggered by a surge of FSH production.
 - c) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
 - d) ischemic phase occurs when the oocyte is fertilized.
 - e) None of the above.

- _____ 8. Concerning the *female reproductive system*,
- a) the usual site of fertilization is the ampulla of the uterine tube.
 - b) the endometrium has a wall made up of a very thick smooth muscle layer.
 - c) the basal layer of endometrium is sloughed off during each menstruation.
 - d) the perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
 - e) the vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

- _____ 9. Which statement is false?
- a) The basal layer of the endometrium is sloughed off during menstruation.
 - b. If a sperm penetrates the secondary oocyte, the second meiotic division is completed.
 - c. The number of oocytes that ovulate is greatly reduced in women who take oral contraceptives.
 - d. At ovulation, the nucleus of the secondary oocyte begins the second meiotic division, but progresses only to metaphase.
 - e. Removal of the corpus luteum of pregnancy before the 4th month usually leads to abortion.

- _____ 10. With respect to the *zona pellucida*, which statement is false?
- a. is composed of sulfated glycoproteins.
 - b. prevents polyspermy
 - c. prevents premature implantation
 - d. functions as a porous filter
 - e. arises by delamination from the embryoblast.

56. Which of the following statements is *true* in the *female reproductive system*?
- a. The cells of theca interna secrete progesterone.

- b. Germinal (surface) epithelium covering the ovary gives rise to ova.
- c. Glassy membrane is the last structure to disappear in an atretic follicle.
- d. The alkaline environment of vagina prevents growth of microorganisms.
- e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.

b

EMBRYOLOGY

MCQ: Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case. (Total : 50 points)

_____ 2. The **heart primordium** between the fourth and seventh weeks **does not** consist of:
a. the bulbus cordis b. ventricle c. atrium d. sinus venosus e. auricles.

_____ 2) The **transverse pericardial sinus** forms from the breaking away of?
a) Somites b) Notocord c) Axial Mesentary d) Dorsal Mesentery e) Ectoderm

_____ 4) What structure does the **proximal third of the bulbus cordis** become?
a) Right Ventricle b) Left Ventricle c) Ventricular Outflow Tracts
d) Aorta & Pulmonary Artery e) Left & Right Atria

_____ 6) What structure does the **distal third (truncus arteriosus) of the bulbus cordis** become?
a) Right Ventricle b) Left Ventricle c) Ventricular Outflow Tracts
d) Aorta & Pulmonary Artery e) Left & Right Atria

_____ 7) What major structure is formed by the **sinus venosus**?
a) Aorta b) Left Ventricle c) Right Ventricle d) Left Atrium e) Right Atrium

_____ 8) Where is the **ostium primum** formed?
a) Between Right and Left Ventricle b) Between Right and Left Atria c) Within the Coronary Sinus
d) Between Right Atrium and Ventricle e) Between Left Atrium and Ventricle

_____ 2. Concerning the **development of the heart**, which statement is **true**?
a) SA nodes are derived from cells from the atrioventricular canal.

- b) the bulbus cordis forms the trabeculated part of the right ventricle.
- c) endocardial cushions develop in the atrioventricular and conotruncal regions.
- d) the prevalence of atrial septal defects is more common in females than in males.
- e) the truncus arteriosus forms the roots and proximal portion of the aorta and pulmonary artery.

_____ 6. Concerning the **development of the heart**, which statement is **false**?

- a) Cavitation of the ventricular walls forms the trabeculae carneae.
- b) The critical period of heart development is from day 20 to day 50 after fertilization
- c) The left horn of sinus venosus becomes incorporated into the wall of the right atrium.
- d) The sinoatrial node develops during the 5th week.
- e) The left auricle is derived from the primordial atrium.

_____ 16) Diethylstilbestrol (DES) was once prescribed to women to prevent miscarriages. However, it has since been found to be a teratogen that causes what type of defects?

- a) Reproductive system
- b) Central nervous system
- c) Gastrointestinal
- d) Limb (partial or total absence)
- e) Virtually any type of defect

_____ 17) Which of the following defects is the most common when the mother has **diabetes**?

- a) Malformation of the uterus
- b) Cleft lip/palate
- c) Heart and neural tube defects
- d) Microcephaly
- e) Cataracts/ Glaucoma

_____ 1. The **cardiovascular system** is **not** derived from:

- a) Splanchnic mesoderm
- b) Endoderm
- c) Paraxial mesoderm
- d) lateral mesoderm
- e) Neural crest cells

_____ 3. With regard to the **development of the arterial system**, which statement is **false**?

- a) the external carotid artery is a sprout of the 4th aortic arch.
- b) the mesenteric arteries are derived from the right proximal umbilical arteries.
- c) the aortic sac forms right and left horns which subsequently gives rise to the brachiocephalic artery.
- d) the 1st aortic arch gives rise to stapedial arteries.
- e) All of the above.
- f) None of the above.

_____ 8. With regard to the **development of cardiovascular system**, which statement is **false**?

- a) The visceral pericardium is derived from mesothelial cells of the bulbus cordis.
- b) The right anterior cardinal vein and the right common cardinal vein become the SVC
- c) The foramen primum serves as a shunt, enabling oxygenated blood to pass from the right to the left atrium.
- d) Because it is derived from the sinus venosus, the smooth part of the wall of the right atrium is called the sinus venarum.
- e) Most of the wall of the left atrium is smooth because it is formed by incorporation of the primordial pulmonary vein.

_____ 5. Concerning the **development of the aortic arches**,

- a) the right fourth aortic arch becomes the right pulmonary artery.
- b) dorsal parts of the fifth pair of aortic arches form the ductus arteriosus.
- c) proximal parts of the third aortic arches persist to form the stems of stapedial arteries.
- d) the first pair of the aortic arches may contribute to the formation of the external carotid arteries.
- e) dorsal parts of the second aortic arches become the proximal part of the right subclavian artery.

_____ 7. With regard to the **development of cardiovascular system**, which statement is **false**?

- a) the foramen ovale functionally closes immediately after birth.
- b) Umbilical veins carry well-oxygenated blood from the chorion.
- c) the smooth part of the right ventricle is derived from the bulbus cordis.
- d) trabeculated part of the right ventricle is derived from the primitive ventricle.
- e) Distal parts of the umbilical arteries become the internal iliac arteries and superior vesical arteries.

_____ 9. Which of the following is not the classic group of cardiac defects of the **Tetralogy of Fallot**?

- a. Pulmonary stenosis (obstruction of right ventricular outflow)
- b. Ventricular septal defect
- c. Dextroposition of aorta (overriding or straddling aorta)
- d. Left ventricular hypertrophy
- e. Right ventricular hypertrophy

_____ 10. Select the **incorrect** statement pertaining to the **development of the aortic arches**.

- a. The left subclavian artery is not derived from any of the pharyngeal arch arteries.
- b. Distal parts of the third pair of pharyngeal arch arteries join to form the internal carotid arteries.
- c. The right fourth pharyngeal arch artery becomes the proximal part of the right subclavian artery.
- d. First pair of pharyngeal arch arteries may contribute to the formation of the external carotid arteries.
- e. Ventral parts of the second pair of pharyngeal arch arteries persist and form the stems of the maxillary arteries.

_____ 11. With regard to the **development of cardiovascular system**, which statement is **false**?

- a. Coarctation of the aorta occurs twice as often in males as in females.
- b. Atrial septal defect (ASD) occurs more frequently in females than in males.
- c. Ventricular septal defects (VSDs) occur more frequently in males than in females.
- d. Persistent ductus arteriosus (PDA) is two to three times more frequent in females than in males.
- e. The distal part of the right sixth pharyngeal arch artery forms a prenatal shunt, the ductus arteriosus.

_____ 12. With regard to the **development of cardiovascular system**, which statement is **false**?

- a. Most of the intra-abdominal parts of the umbilical arteries become the ligamentum venosum.
- b. The short, thick ligamentum arteriosum extends from the left pulmonary artery to the arch of the aorta.
- c. The right ventricular wall is thicker than the left ventricular wall in fetuses and newborn infants.
- d. The intra-abdominal part of the umbilical vein eventually becomes the round ligament of liver.
- e. None of the above.

_____ 9) What is the correct order of **cardiac conduction**?

- a) AV node to SA node to His bundle to Purkinje fibers
- b) Purkinje fibers to His bundle to SA node to AV node
- c) SA node to AV node to His bundle to Purkinje fibers
- d) Purkinje fibers to His bundle to AV node to SA node
- e) Bachman's bundle to SA node to AV node to ventricles

_____ 10) Which of the following is **NOT** associated with an **Atrial Septal Defect**?

- a) Persistent ostium secundum
- b) Endocardial cushion defect
- c) Sinus venosum defect
- d) Probe patency
- e) Transposition of the Great Vessels

_____ 4) Which nerve is associated with the aortic arches IV and VI?

- a) V – Trigeminal Nerve
- b) VI – Abducent Nerve
- c) IX – Glossopharyngeal Nerve
- d) X – Vagus Nerve
- e) XI – Accessory Nerve

_____ 5) Aortic arch IV forms the _____ on the right side of the embryo and the _____ on the left.

- a) Right subclavian artery; Arch of aorta
- b) Ductus arteriosus; Pulmonary Artery

- c) Arch of aorta; Pulmonary Artery
e) Right subclavian artery; Left subclavian artery

d) Ductus arteriosus; Right subclavian artery

_____ 8) In utero, the ductus venosus helps shunt blood away from the very first organ it reaches to more important organs like the brain. This shunt bypasses the _____.

- a) Lungs b) Aorta c) Spleen d) Pancreas e) Liver

_____ 9) During fetal life, in which of the following structures is the percent hemoglobin/oxygen saturation level of fetal blood the lowest?

- a) Right Atrium b) Inferior Vena Cava c) Umbilical Vein
d) Ductus Arteriosus e) Descending Aorta

_____ 10) During week six (lymphatic vessel development), the dilated lymph sac that forms at the lower end of the thoracic duct is called the _____.

- a) Lymphocitic Sac b) Iliac Lymph Sac c) Retroperitoneal Lymph Sac
d) Jugular Lymph Sac e) Cisterna Chyli

_____ 30. **Vitelline arteries** give rise to all of the following arteries, **except**:

- a) Celiac artery b) Renal artery c) Superior mesenteric artery d) Inferior mesenteric artery d)
None

_____ 33. Which of the following primitive veins gives the **hepatic segment of inferior vena cava**?

- a) Right supracardinal vein b) Right subcardinal vein c) Right vitelline vein
d) Right Umbilical vein e) Right posterior cardinal vein

_____ 34. Which of the following vessels carries relatively the most **deoxygenated blood in the fetal blood circulation**?

- a) Descending aorta b) Inferior vena cava c) Umbilical vein
d) Ascending aorta e) Ductus venosus

_____ 35. After birth **umbilical vein** becomes:

- a) Medial umbilical ligament b) Median umbilical ligament c) Lateral umbilical ligament
d) Ligamentum teres hepatis e) Ligamentum venosum

-----19. With regard to the **development of the venous system**, which statement is **true**?

- a) the anastomosis between the anterior cardinal veins forms the left renal vein.
b) the anastomosis between the subcardinal veins forms the left brachiocephalic vein.
c) the anastomosis between the sacrocardinal veins forms the left common iliac vein.
d) the left subcardinal vein forms the azygos vein. e) none of the above

_____ 21. With regard to the **congenital anomalies of the kidneys and ureters** which statement is **false**?

- a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

_____ 22. Which of the following is not the **adult derivative of the ureteric bud**?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ **24. Urachal cyst** is a remnant of:

- a) urogenital sinus b) urogenital rigde c) cloaca d) allantois e) mesonephric duct

_____ **25. The podocytes of Bowman's capsule** are derived from:

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ **26. A structure found within the adult female pelvis formed from the gubernaculum** is

- a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ **47. Concerning the development of the urinary system, which statement is false?**

- a) Urinary system begins to develop before the genital system.
b) The metanephric diverticulum is the primordium of the ureter.
c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
d) Differentiation of the nephrons depends upon induction by the collecting tubules.
e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **48. Concerning the development of the urinary bladder and urethra, which statement is false?**

- a) The female urethra is derived from endoderm of the urogenital sinus.
b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ **49. With regard to the congenital anomalies of the kidneys and ureters which statement is false?**

- a) Accessory renal arteries are end arteries.
b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

_____ **50. During the development of the Urinary System, which statement is true?**

- a) The urorectal septum divides the urachus from the allantois.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) The podocytes of Bowman's capsule are derived from endoderm and ectoderm.
d) Urachal cyst is a remnant of the cloaca. e) Urine production begins shortly after birth.

_____ **78. Which of the following is not the adult derivative of the ureteric bud?**

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ **79. When does the metanephros become functional?**

- a) at week 3 of development b) at week 4 of development
c) at week 10 of development d) just before birth e) just after birth

____ 80. Urachal cyst is a remnant of
a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e) mesonephric duct

____ 81. The transitional epithelium lining the urethra is derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

____ 82. The podocytes of Bowman's Capsule are derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

____ 83. The proximal convoluted tubules of the definitive adult kidney are derived from
a) ureteric bud b) metanephric vesicle c) metanephric duct
d) pronephric tubules e) metanephric tubules

____ 84. A structure found within the adult female pelvis formed from the gubernaculum is
a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

____ **40.** Concerning the **development of the urinary System**, which statement is **false**?
a) Urinary system begins to develop before the genital system
b) The metanephric diverticulum is the primordium of the ureter.
c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
d) Differentiation of the nephrons depends upon induction by the collecting tubules.
e) The kidneys attain their definitive adult position toward the end of the fetal period.

____ **41.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?
a) The female urethra is derived from endoderm of the urogenital sinus.
b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

____ **5.** With regard to the **congenital anomalies of the kidneys and ureters** which statement is **false**?
a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

____ **6.** Which of the following is not the **adult derivative of the ureteric bud**?
a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

____ **7. Urachal cyst** is a remnant of:

- a) urogenital sinus b) urogenital ridge c) cloaca d) allantois e)
mesonephric duct

_____ 8. The **podocytes of Bowman's capsule** are derived from:

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ 17. Concerning the **development of the Urinary System**, which statement is **false**?

- a) The kidneys attain their definitive adult position postnatally.
b) Urinary system begins to develop before the genital system.
c) The metanephric diverticulum is the primordium of the ureter.
d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
e) Differentiation of the nephrons depends upon induction by the collecting tubules.

_____ 18. Which of the following is not the adult derivative of the **ureteric bud**?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ 19. Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ 40. During the **pronephros period** (week 3), the first kidney develops in the _____ region.

- a) Cervical b) Thoracic c) Lumbar d) Sacral e) Cloaca

_____ 41. The **ureteric bud** gives rise to the _____ and the **metanephric mesoderm** gives rise to the _____.

- a) Nephrons; Collecting system b) Collecting system; Nephrons c) Bowman's capsule; Ureter
d) Loop of Henle; Renal pelvis e) Distal/proximal convoluted tubules; Major/minor calyces

_____ 42. The **urorectal septum** divides the _____ from the _____.

- a) Anorectal canal; Urogenital sinus b) Urogenital sinus; Mesonephric duct c) Urachus; Allantois
d) Urinary bladder; Allantois e) Mesonephric duct; Ureteric bud

_____ 43. The **urachus** forms the:

- a) Allantois b) Median umbilical ligament c) Medial umbilical ligament
d) Lateral umbilical ligament e) Round (umbilical) ligament

_____ 44. The **bladder** becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.

- a) Urethra b) Prostate c) Trigone d) Posterior wall e) Anterior wall

_____ 11. During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
- b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
- d) Urine production begins shortly after birth.
- e) At birth, the kidneys have a lobulated appearance.

_____ **55.** The ***transitional epithelium lining the ureter*** is derived from
 a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ **56.** Which of the following is not the adult derivative of the ***ureteric bud***?
 a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ **59.** With regard to the **congenital anomalies of the kidneys** and ureters which statement is **false**?

- a) Accessory renal arteries are end arteries.
- b) Bilateral renal agenesis is associated with oligohydramnios.
- c) Incontinence is the common complaint resulting from an ectopic ureteric orifice in both sexes.
- d) Unilateral renal agenesis is more common in males than females.
- e) None of the above.

_____ **58.** Concerning the **development of the Urinary System**, which statement is **false**?

- a) Urinary system begins to develop before the genital system
- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **60.** Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- b) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
- c) In adults, the urachus is represented by the median umbilical ligament.
- d) The urinary bladder becomes a pelvic organ during the 9th week of development.
- e) The female urethra is derived from endoderm of the urogenital sinus.

_____ **63.** During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.

b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.

c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.

d) Urine production begins shortly after birth.

e) At birth, the kidneys have a lobulated appearance.

_____ **44.** With regard to the **development of gonads**, which statement is **false?**

a) The surface epithelium (germinal epithelium) of the female gonad develop into oogonia.

b) Sex of an embryo is determined genetically at the time of fertilization.

c) Germ cells do not appear in the genital ridges until the 6th week of development.

d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.

e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **45.** With regard to the **vestigial structures derived from embryonic genital system**, which statement is **false?**

a) A structure found within the adult female pelvis formed from the gubernaculum is median umbilical ligament

b) The cranial end of mesonephric duct in males may persist as paradidymis.

c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.

d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.

e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **21.** During the development of the **female reproductive system**, which statement is **false?**

a) The urogenital folds remain unfused and form the labia minora.

b) The cranial portion of the paramesonephric ducts form the uterine tubes.

c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.

d) The phallus forms the mons pubis.

e) The labioscrotal swellings form the labia majora.

_____ **22.** Concerning the **development of the genital system**, one statement is **false?**

a) Bilateral cryptorchidism may result in sterility.

b) Hypospadias is constantly associated with exstrophy of urinary bladder.

c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.

d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora

e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.

_____ **23.** With regard to the **development of gonads**, which statement is **false?**

a) The surface epithelium of the female gonad develop into oogonia.

b) Sex of an embryo is determined genetically at the time of fertilization.

c) Germ cells do not appear in the genital ridges until the 6th week of development.

d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.

e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **24.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false?**

a) The prostatic utericle is homologous to the vagina.

b) The cranial end of mesonephric duct in males may persist as paradidymis.

- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
 d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
 e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **10.** A structure found within the adult **female pelvis formed from the gubernaculum** is
 a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ **12.** The **labia minora** arise embryologically from:
 a) phallus b) labioscrotal swellings c) sinovaginal bulbs d) urogenital folds e)
 paramesonephric duct

_____ **13.** The **uterine tubes** of the adult female are derived embryologically from
 a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
 d) paramesonephric tubules e) uterovaginal primordium

_____ **14.** The most common cause of **testicular feminization** [Androgen Insensitivity] **syndrome** is
 a) a 46, XO genotype b) a 47, XXY genotype c) lack of androgen receptors
 d) congenital adrenal hyperplasia e) inadequate production of testosterone and müllerian-inhibiting factor.

_____ **15.** In the male, **failure of urethral folds to fuse completely** results in
a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

_____ **46.** During the development of the **female reproductive system**, which statement is **false**?
 a) The urogenital folds remain unfused and form the labia minora.
 b) The cranial portion of the paramesonephric ducts form the uterine tubes.
 c) During development of the ovaries, the primitive sex (medullary) cords degenerate.
 d) The phallus forms the mons pubis. e) The labioscrotal swellings form the labia majora.

_____ **47.** During the development of the **male reproductive system**, which statement is **false**?
 a) Hypospadias is constantly associated with exstrophy of urinary bladder.
 b) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
 c) Sertoli cells, derived from mesoderm, constitute most of the seminiferous epithelium in the fetal testis.
 d) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
 e) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the paramesonephric ducts.

_____ **49.** A patient is born with karyotype **46, XY** but has no problem with production of testosterone or müllerian inhibiting hormone. Their **vagina ends in a blind pouch** and **testes are lodged in the inguinal canal**. Which of the following is the best description of this patient:
 a) Male pseudo-hermaphrodite b) Female pseudo-hermaphrodite c) Androgen insensitivity syndrome

d) Turner's syndrome

e) Klinefelter syndrome

_____ **50.** During development of the **ovaries, the primitive sex (medullary) cords**

- a) Proliferate b) Move laterally c) Move medially d) Differentiate e) Degenerate

_____ **51.** The **gubernaculum** is attached to the uterus and as the ovaries descend it becomes the _____ of the uterus.

- a) Canal of Nuck b) Ovarian ligament c) Round ligament
d) Ovarian and round ligaments e) Gubernacular ligament

_____ **52. Development of the testis** is induced by the **sry-gene**, which encodes testis-determining factor (TDF). TDF causes all of the following **EXCEPT**:

- a) The sex cords continue to proliferate to form the testis.
b) Leydig cells are secreting testosterone thus have very little SER.
c) Towards the hilum, the cords grow into a network that will give rise to the rete testes.
d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium.
e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

_____ **53.** Which of the following inhibits the growth of the **paramesonephric duct (males)?**

- a) Testosterone b) Leydig Cells c) Dihydrotestosterone d) Estrogens e) Müllerian substances (MIFs)

_____ **54.** As the **testes descend**, they follow the:

- a) Processus vaginalis b) Parietal layer of tunica vaginalis c) Visceral layer of tunica vaginalis
d) Caudal genal ligament e) Cryptorchidism

_____ **55.** The **epididymis, ductus deferens, ejaculatory duct, and seminal vesicle** all arise from:

- a) The Müllerian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Wolffian hormones

_____ **56.** The **uterine tubes, uterus, cervix, and superior part (1/3) of the vagina** all arise from:

- a) The Wolffian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Müllerian hormones

_____ **57.** The **epioophoron and paroophoron**, which are important as cysts, arise from:

- a) The Mullarian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Wolffian hormones

_____ **58.** The **lower portion of the vagina** is derived from:

- a) Sinovaginal bulb b) Uteral lumen c) Hymen d) Urogenital sinus e) Vaginal plate

- b) The uterine tubes of the adult female are derived embryologically from uterovaginal primordium.
- c) The most common cause of testicular feminization syndrome is the presence of abundant androgen receptors.
- d) Round ligament of the uterus is formed from the gubernaculum.
- e) The X chromosome carries a gene on its short arm that codes for TDF.

_____ **71.** During the **development of the testes**, which statement is **false**?

- a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.
- b) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- c) The fetal testis produces antimüllerian hormone by Sertoli cells at 6-7 weeks
- d) Sertoli cells are derived from the surface epithelium of the testis.
- e) None of the above.

_____ **69.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The cranial end of mesonephric duct in males may persist as paradidymis.
- b) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- c) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- d) The prostatic utricle is homologous to the vagina.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **70.** Concerning the **development of the genital system**, which statement is **false**?

- a) Inguinal canals develop in both sexes.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- d) Gubernaculum does not play any role in the descent of ovaries.
- e) The persistent processus vaginalis in the female is called canal of Nuck.

_____ **72.** With regard to **development of the ovaries**, which statement is **false**?

- a) The cranial part of the gubernaculum becomes the ovarian ligament.

- b) As the ovary separates from the regressing mesonephros, it is suspended by mesovarium.
- c) The surface epithelium becomes separated from the follicles in the cortex by tunica albuginea.
- d) The primordial follicles are derived from the rete ovarii. e) No oogonia are formed postnatally.

____ 85. After the sinovaginal bulbs have proliferated and fused, they form a solid core of endodermal cells called

- a) vestibule of the vagina b) uterovaginal primordium c) urogenital sinus
d) vaginal plate e) clitoris

____ 86. The labia minora arise embryologically from

- a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct

____ 87. The uterine tubes of the adult female are derived embryologically from

- a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

____ 88. The most common cause of female pseudointersexuality is

- a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 89. The most common cause of male pseudointersexuality is

- a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 90. The most common cause of testicular feminization syndrome is

- a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 91. In the male, failure of urethral folds to fuse completely results in

- a) hypospadias b) epispadias c) cryptorchidism
d) hydrocele e) congenital inguinal hernia

____ 92. The Y chromosome carries a gene on its short arm that codes for

- a) testosterone b) MIF c) testes-determining factor (TDF) d) progesterone e) estrogen

102. The process of gastrulation is first indicated by the formation of the

- a) prochordal plate b) cloacal membrane c) primitive streak d) neural tube e) somites

32. Concerning the development of the genital system, which statement is false?

- a) Inguinal canals develop in both sexes.
b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.

- d) Gubernaculum does not play any role in the descent of ovaries.
e) The persistent processus vaginalis in the female is called canal of nuck.

33. In the male, **failure of urethral folds to fuse** completely results in
a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

11. During the development of the testes, which statement is false?
a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.
b) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
c) The fetal testis produces antimüllerian hormone by Sertoli cells at 6-7 weeks
d) Sertoli cells are derived from the surface epithelium of the testis.
e)

The labia minora arise embryologically from paramesonephric ducts.fff

The uterine tubes of the adult female are derived embryologically from uterovaginal primordium.fff

The most common cause of testicular feminization syndrome is lack of androgen receptors.ttt

Round ligament of the uterus is formed from the gubernaculum. ttt

The Y chromosome carries a gene on its short arm that codes for testes-determining factor (TDF).ttt

- ____ 81. The transitional epithelium lining the urethra is derived from
a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

- ____ 83. The proximal convoluted tubules of the definitive adult kidney are derived from
a) ureteric bud b) metanephric vesicle c) metanephric duct
d) pronephric tubules e) metanephric tubules

- ____ 84. A structure found within the adult female pelvis formed from the gubernaculum is
a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

85. After the sinovaginal bulbs have proliferated and fused, they form a solid core of endodermal cells called

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d) vaginal plate e) clitoris

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a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct

- ____ 87. The uterine tubes of the adult female are derived embryologically from
a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

- ____ 88. The most common cause of female pseudointersexuality is
a) a 46, XO genotype b) a 47, XXY genotype

- c) lack of androgen receptors
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)
- d) congenital adrenal hyperplasia

____ 89. The most common cause of male pseudointersexuality is

- a) a 46, XO genotype
c) lack of androgen receptors
hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)
- b) a 47, XXY genotype
d) congenital adrenal

____ 90. The most common cause of testicular feminization syndrome is

- a) a 46, XO genotype
c) lack of androgen receptors
hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)
- b) a 47, XXY genotype
d) congenital adrenal

____ 91. In the male, failure of urethral folds to fuse completely results in

- a) hypospadias
d) hydrocele
- b) epispadias
e) congenital inguinal hernia
- c) cryptorchidism

____ 92. The Y chromosome carries a gene on its short arm that codes for

- a) testosterone b) MIF c) testes-determining factor (TDF) d) progesterone e) estrogen

____ 78. Which of the following is not the adult derivative of the ureteric bud?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

____ 79. When does the metanephros become functional?

- a) at week 3 of development
c) at week 10 of development
birth
- b) at week 4 of development
d) just before birth
e) just after

____ 83. The proximal convoluted tubules of the definitive adult kidney are derived from

- a) ureteric bud
d) pronephric tubules
- b) metanephric vesicle
e) metanephric tubules
- c) metanephric duct

____ 84. A structure found within the adult female pelvis formed from the gubernaculum is

- a) broad ligament
d) round ligament of the uterus
- b) superior ligament of the ovary
e) median umbilical ligament
- c) medial umbilical ligament

____ 85. After the sinovaginal bulbs have proliferated and fused, they form a solid core of endodermal cells called

- a) vestibule of the vagina
d) vaginal plate
- b) uterovaginal primordium
e) clitoris
- c) urogenital sinus

____ 86. The labia minora arise embryologically from

- a) phallus
bulbs
d) urogenital folds
- b) labioscrotal swellings
e) paramesonephric duct
- c) sinovaginal

____ 87. The uterine tubes of the adult female are derived embryologically from

- a) mesonephric duct
paramesonephric duct
- b) mesonephric tubules
c) paramesonephric duct

d) paramesonephric tubules

e) uterovaginal primordium

____ 88. The most common cause of female pseudointersexuality is

a) a 46, XO genotype

b) a 47, XXY genotype

c) lack of androgen receptors

d) congenital adrenal

hyperplasia

e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 89. The most common cause of male pseudointersexuality is

a) a 46, XO genotype

b) a 47, XXY genotype

c) lack of androgen receptors

d) congenital adrenal

hyperplasia

e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 90. The most common cause of testicular feminization syndrome is

a) a 46, XO genotype

b) a 47, XXY genotype

c) lack of androgen receptors

d) congenital adrenal

hyperplasia

e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

____ 91. In the male, failure of urethral folds to fuse completely results in

a) hypospadias

b) epispadias

c) cryptorchidism

d) hydrocele

e) congenital inguinal hernia

____ 92. The Y chromosome carries a gene on its short arm that codes for

a) testosterone

b) MIF

c) testes-determining factor (TDF)

d) progesterone

e) estrogen

21. During the urinary system,

-----a) the metanephros begins to form at week 5.

-----b) if the ureteric bud undergoes division prematurely, duplication of the ureter and renal pelvis occurs.

-----c) the urinary bladder develops from the upper end of the urogenital sinus.

-----d) the ureter develops from the ureteric bud .

-----e) the ureteric bud is a diverticulum from the mesonephric duct.

-----f) the podocytes of Bowman's capsule develop from the metanephric vesicles.

-----a) the transitional epithelium lining the entire urinary bladder is of endodermal origin.

____ **27.** The **labia minora** arise embryologically from:

a) phallus

b) labioscrotal swellings

c) sinovaginal bulbs

d) urogenital folds

e) paramesonephric duct

____ **28.** The **uterine tubes** of the adult female are derived embryologically from

a) mesonephric duct

b) mesonephric tubules

c)

paramesonephric duct

d) paramesonephric tubules

e) uterovaginal primordium

____ **29.** The most common cause of **testicular feminization syndrome** is

a) a 46, XO genotype

b) a 47, XXY genotype

c) lack of androgen receptors

d) congenital adrenal hyperplasia

e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

_____ **30.** In the male, ***failure of urethral folds to fuse completely*** results in
a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital
inguinal hernia

_____ **32. *Intermediate medoderm*** gives rise to the
a) neural tube b) heart c) kidneys and gonads d) somites e)
notochord

_____ **36.** Concerning the ***development of the Urinary System***, which statement is ***false?***
a) The kidneys attain their definitive adult position postnatally.
b) Urinary system begins to develop before the genital system.
c) The metanephric diverticulum is the primordium of the ureter.
d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
e) Differentiation of the nephrons depends upon induction by the collecting tubules.

_____ **38.** Which of the following is not the adult derivative of the ***ureteric bud?***
a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting
tubule

_____ **39.** Concerning the ***development of the urinary bladder and urethra***, which statement is
false?
a) The female urethra is derived from endoderm of the urogenital sinus.
b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the
urogenital sinus.

_____ **40.** During the ***development of the kidney systems***, which statement is ***true?***
a) The transitional epithelium lining the urethra is derived from neuroectoderm.
b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar
segments.
d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated
appearance.

_____ **41.** During the development of the ***female reproductive system***, which statement is ***false?***
a) The urogenital folds remain unfused and form the labia minora.
b) The cranial portion of the paramesonephric ducts form the uterine tubes.
c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric
ducts.
d) The phallus forms the mons pubis. e) The labioscrotal swellings form the labia
majora.

_____ **42.** Concerning the ***development of the genital system***, one statement is ***false?***
a) Bilateral cryptorchidism may result in sterility.

- b) Hypospadias is constantly associated with exstrophy of urinary bladder.
- c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.
- e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.

_____ **43.** With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) Germ cells do not appear in the genital ridges until the 6th week of development.
- d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **44.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The prostatic utericle is homologous to the vagina.
- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **11.** During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
- b) The “ascent” of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
- d) Urine production begins shortly after birth.
- e) At birth, the kidneys have a lobulated appearance.

_____ **12.** The **labia minora** arise embryologically from:

- a) phallus b) labioscrotal swellings c) sinovaginal bulbs d) urogenital folds e) paramesonephric duct

_____ **13.** The **uterine tubes** of the adult female are derived embryologically from

- a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
- d) paramesonephric tubules e) uterovaginal primordium

_____ **14.** The most common cause of **testicular feminization** [Androgen Insensitivity] **syndrome** is

- a) a 46, XO genotype b) a 47, XXY genotype c) lack of androgen receptors
- d) congenital adrenal hyperplasia e) inadequate production of testosterone and müllerian-inhibiting factor.

_____ **15.** In the male, ***failure of urethral folds to fuse completely*** results in
a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal
hernia

_____ **16.** ***Intermediate mesoderm*** gives rise to the
a) neural tube b) heart c) kidneys and gonads d) somites e)
notochord

_____ **17.** Concerning the **development of the Urinary System**, which statement is **false**?
a) The kidneys attain their definitive adult position postnatally.
b) Urinary system begins to develop before the genital system.
c) The metanephric diverticulum is the primordium of the ureter.
d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
e) Differentiation of the nephrons depends upon induction by the collecting tubules.

_____ **18.** Which of the following is not the adult derivative of the ***ureteric bud***?
a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting
tubule

_____ **19.** Concerning the **development of the urinary bladder and urethra**, which statement is
false?
a) The female urethra is derived from endoderm of the urogenital sinus.
b) In adults, the urachus is represented by the median umbilical ligament.
c) The urinary bladder becomes a pelvic organ during the 9th week of development.
d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital
sinus.

_____ **20.** Up to the 5th week of development, the embryonic **skin** is only a single layer of:
a) Ectoderm b) Endoderm c) Paraxial
mesoderm d) Intermediate mesoderm e) Lateral plate mesoderm

_____ **21.** During the development of the **female reproductive system**, which statement is **false**?
a) The urogenital folds remain unfused and form the labia minora.
b) The cranial portion of the paramesonephric ducts form the uterine tubes.
c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of
paramesonephric ducts.
d) The phallus forms the mons pubis. e) The labioscrotal swellings form the labia majora.
Cowper's glands [bulbourethral glands] in the male is homologous to the greater vestibular
gland in the female.

_____ **22.** Concerning the **development of the genital system**, one statement is **false**?
a) Bilateral cryptorchidism may result in sterility.
b) Hypospadias is constantly associated with exstrophy of urinary bladder.
c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora
e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as
chordee.

_____ **23.** With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) Germ cells do not appear in the genital ridges until the 6th week of development.
- d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **24.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The prostatic utericle is homologous to the vagina.
- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **41.** The **ureteric bud** gives rise to the _____ and the **metanephric mesoderm** gives rise to the _____.

- a) Nephrons; Collecting system
- b) Collecting system; Nephrons
- c) Bowman's capsule; Ureter
- d) Loop of Henle; Renal pelvis
- e) Distal/proximal convoluted tubules; Major/minor calyces

_____ **42.** The **urorectal septum** divides the _____ from the _____.

- a) Anorectal canal; Urogenital sinus
- b) Urogenital sinus; Mesonephric duct
- c) Urachus; Allantois
- d) Urinary bladder; Allantois
- e) Mesonephric duct; Ureteric bud

_____ **43.** The **urachus** forms the:

- a) Allantois
- b) Median umbilical ligament
- c) Medial umbilical ligament
- d) Lateral umbilical ligament
- e) Round (umbilical) ligament

_____ **44.** The **bladder** becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.

- a) Urethra
- b) Prostate
- c) Trigone
- d) Posterior wall
- e) Anterior wall

_____ **45.** At birth, a child has **large palpable flank masses and abnormal extremities**. The delivery physician is very concerned as she knows this **autosomally-dominant disorder** can lead to renal failure and the child will likely need a kidney transplant to survive. Which of the following disorders does the child likely have?

- a) Polycystic kidney
- b) Renal agenesis
- c) Pelvic kidney
- d) Bifid ureter
- e) Bifid kidney

_____ **46.** This disorder, associated with **oligohydramnios**, occurs when the ureteric bud fails to reach the metanephric mesoderm.

- a) Polycystic kidney
- b) Renal agenesis
- c) Pelvic kidney
- d) Horseshoe kidney
- e) Bifid ureter

_____ **47.** What artery prevents the **horseshoe kidney** (metanephric mesodermal fusion) from ascending high into the abdomen?

- a) Celiac trunk
- b) Superior mesenteric artery
- c) Inferior mesenteric artery
- d) Left renal artery
- e) Right renal artery

_____ **48.** A newborn baby presents with fluid draining from the **umbilicus onto the skin**. Testing of the fluid identifies it as urine. What is the most likely diagnosis?

- a) Urachal cyst
- b) Urachal fistula
- c) Exstrophy of the bladder
- d) Bifid ureter
- e) Horseshoe kidney

_____ **49.** A patient is born with karyotype **46, XY** but has no problem with production of testosterone or müllerian inhibiting hormone. Their **vagina ends in a blind pouch** and **testes are lodged in the inguinal canal**. Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite
- b) Female pseudo-hermaphrodite
- c) Turner's syndrome
- d) Androgen insensitivity syndrome
- e) Klinefelter syndrome

_____ **50.** During development of the **ovaries, the primitive sex (medullary) cords** _____.

- a) Proliferate
- b) Move laterally
- c) Move medially
- d) Differentiate
- e) Degenerate

_____ **51.** The **gubernaculum** is attached to the uterus and as the ovaries descend it becomes the _____ of the uterus.

- a) Canal of Nuck
- b) Ovarian ligament
- c) Round ligament
- d) Ovarian and round ligaments
- e) Gubernacular ligament

_____ **52. Development of the testis** is induced by the sry-gene, which encodes testis-determining factor

(TDF). TDF causes all of the following **EXCEPT**:

- a) The sex cords continue to proliferate to form the testis.
- b) Leydig cells are secreting testosterone thus have very little SER.
- c) Towards the hilum, the cords grow into a network that will give rise to the rete testes.
- d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium.
- e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

_____ **53.** Which of the following inhibits the growth of the **paramesonephric duct (males)?**

- a) Testosterone
- b) Leydig Cells
- c) Dihydrotestosterone
- d) Estrogens
- e) Müllerian substances (MIFs)

_____ **54.** As the **testes descend**, they follow the:

- a) Processus vaginalis
- b) Parietal layer of tunica vaginalis
- c) Visceral layer of tunica vaginalis
- d) Caudal genal ligament
- e) Cryptorchidism

_____ **55.** The **epididymis, ductus deferens, ejaculatory duct, and seminal vesicle** all arise from:

- a) The Müllerian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

_____ **56. The uterine tubes, uterus, cervix, and superior part (1/3) of the vagina** all arise from:

- a) The Wolffian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Müllerian hormones

_____ **57. The epoophoron and paroophoron**, which are important as cysts, arise from:

- a) The Mullarian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Wolffian hormones

_____ **58. The lower portion of the vagina** is derived from:

- a) Sinovaginal bulb b) Uteral lumen c) Hymen d) Urogenital sinus e)
Vaginal plate

_____ **59. What structure in the indifferent stage** (weeks 3-6) becomes the **clitoris**?

- a) Urogenital groove b) Genital swelling c) Urethral fold d) Perineum e) Genital tubercle

_____ **60. Genetic 46,XY** would be:

- a) Turner's syndrome b) Klinefelter syndrome c) True hermaphrodite
d) Male pseudo-hermaphrodite e) Female pseudo-hermaphrodite

Embryology #13 – Urinary System

1) The urinary system develops mainly from:

- a) Ectoderm b) Paraxial mesoderm c) Intermediate mesoderm d) Lateral plate mesoderm
e) Endoderm

2) During the pronephros period (week 3), the first kidney develops in the ____ region. By week 4 this kidney degenerates.

- a) Cervical b) Thoracic c) Lumbar d) Sacral e) Cloaca

3) The mesonephros (embryologic kidney) forms from weeks 4-8 in the ____ region.

- a) Cervical

b) Thoracic

c) Lumbar

d) Sacral

e) Cloaca

4) The metanephros (weeks 5 and on) give rise to the ureteric bud and form in the ____ region.

a) Cervical

b) Thoracic

c) Lumbar

d) Sacral

e) Cloaca

5) The ureteric bud gives rise to the ____ and the metanephric mesoderm (blastema) gives rise to the ____.

a) Nephrons; Collecting system
Ureter

b) Collecting system; Nephrons

c) Bowman's capsule;

d) Loop of Henle; Renal pelvis

e) Distal/proximal convoluted tubules; Major/minor calyces

6) As the kidney ascends, it rotates from ____-facing to ____-facing (relating to the hilum facing direction).

a) Anterior; Inferior
Medial

b) Posterior; Lateral

c) Anterior; Lateral

d) Posterior;

e) Anterior; Medial

7) The urorectal septum divides the ____ from the ____.

a) Anorectal canal; Urogenital sinus b) Urogenital sinus; Mesonephric duct c) Urinary bladder;
Allantois

d) Urachus; Allantois

e) Mesonephric duct; Ureteric bud

8) The urachus forms the:

a) Allantois

b) Median umbilical ligament

c) Medial umbilical ligament

d) Lateral umbilical ligament

e) Round (umbilical) ligament

9) The urogenital sinus is made from what type of tissue?

a) Muscle

b) Nerve

c) Connective

d) Epithelial

10) The bladder becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.

a) Urethra

b) Prostate

c) Trigone

d) Posterior wall

e) Anterior wall

12) This disorder, associated with oligohydramnios, occurs when the ureteric bud fails to reach the metanephric mesoderm.

a) Polycystic kidney

b) Renal agenesis

c) Pelvic kidney

d) Horseshoe kidney

e) Bifid ureter

13) What artery prevents the horseshoe kidney (metanephric mesodermal fusion) from ascending high into the abdomen?

a) Celiac trunk

b) Superior mesenteric artery

c) Inferior mesenteric

artery

d) Left renal artery

e) Right renal artery

14) Which of the following disorders would show an increased level of alpha fetoprotein (AFP) during maternal Quad testing/screening?

a) Renal agenesis

b) Horseshoe kidney

c) Bifid ureter

d) Ectopic ureter

e) Exstrophy of the bladder

15) A newborn baby presents with fluid draining from the umbilicus onto the skin. Testing of the fluid identifies it as urine. What is the most likely diagnosis?

a) Urachal cyst

b) Urachal fistula

c) Exstrophy of the bladder

d) Bifid ureter

e) Horseshoe kidney

Embryology #14 – Reproductive System

1) Testis determining factor (TDF), a transcription factor, is turned on by a(n) ____ chromosome from the ____.

a) X; egg

b) Y; egg

c) X; sperm

d) Y; sperm

2) The indifferent gonad forms as a pair of longitudinal ridges ____ to the mesonephric ridges.

a) Medial

b) Lateral

c) Superior

d) Inferior

4) All of the following contribute to the indifferent gonad EXCEPT:

a) Ectodermic placodes

b) Endodermic germ cells

c) Coelomic epithelium

d) Mesenchymal cells

5) During development of the ovaries, the primitive sex (medullary) cords ____.

- a) Proliferate b) Move laterally c) Move medially d) Differentiate e) Degenerate

6) The gubernaculum is attached to the uterus and as the ovaries descend it becomes the ____ of the uterus.

- a) Canal of Nuck b) Ovarian ligament c) Round ligament d) Ovarian and round ligaments

e) Gubernacular ligament

7) Development of the testis is induced by the sry-gene, which encodes testis-determiningfactor (TDF). TDF causes all of the following EXCEPT:

a) The sex cords continue to proliferate to form the testis

b) Leydig cells are secreting testosterone thus have very little SER

c) Towards the hilum, the cords grow into a network that will give rise to the rete testes

d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium

e) Sex cords become seminiferous cords that form seminiferous tubules (Sertolicells), straight tubules, and rete testes

8) Which of the following inhibits the growth of the paramesonephric duct (males)?

a) Testosterone

b) Leydig Cells

c) Dihydrotestosterone

d) Estrogens

e) Mullerian substances (MIFs)

9) As the testes descend, they follow the:

a) Processus vaginalis

b) Parietal layer of tunica vaginalis

- c) Visceral layer of tunica vaginalis
- d) Caudal genical ligament
- e) Cryptorchidism

10) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from:

- a) The Mullarian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

11) The uterine tubes, uterus, cervix, and superior part (1/3) of the vagina all arise from:

- a) The Wolffian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Mullarian hormones

12) The epoophoron and paroophoron, which are important as cysts may form from these remnants, arise from:

- a) The Mullarian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

13) The lower portion of the vagina is derived from:

- a) Sinovaginal bulb
- b) Uteral lumen

c) Hymn

d) Urogenital sinus

e) Vaginal plate

14) What structure in the indifferent stage (weeks 3-6) becomes the clitoris?

a) Urogenital groove

b) Genital swelling

c) Urethral fold

d) Perineum

e) Genital tubercle

15) What structure in the indifferent stage (weeks 3-6) becomes the phallus?

a) Urogenital groove
tubercle

b) Genital swelling

c) Urethral fold

d) Perineum

e) Genital

16) Genetic 46XY would be:

a) Turner's syndrome

b) Klinefelter syndrome

c) True hermaphrodite

d) Male pseudo-hermaphrodite

e) Female pseudo-hermaphrodite

17) A child is born with a tumor on their adrenal glands. This leads to adrenal hyperplasia. They display 46XX phenotypical characteristics. Which of the following is the best description of this patient:

a) Turner's syndrome

b) Klinefelter syndrome

c) True hermaphrodite

d) Male pseudo-hermaphrodite

e) Female pseudo-hermaphrodite

19) A young boy is learning how to use a standup urinal and must hold his penis pointing superiorly due to a birth defect. Which of the following describes his defect?

- a) Bifid penis b) Hypospadias c) Hydrocele d) Cryptorchidism e) Bicornis

7) Mammary glands are formed from which of the following?

- a) Neural crest
b) Endoderm
c) Mesenchyme
d) Ectoderm